

**TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS
INCORPORATED (TELTECH)**

NATIONWIDE 5G FIXED WIRELESS ACCESS (FWA) INFRASTRUCTURE PROGRAM

Tender No: RFP-TELTECH-FWA-
2026-001

Author & Lead Architect: OLIVER
TUNGOL

Date: October 6, 2026 · Rev 2.0
(Official Bidding Version)

**APPENDIX B: COMMERCIAL PROPOSAL, BILL OF QUANTITIES
(BOQ) & PRICING TEMPLATES**

5G FWA PROGRAM (5,000 SITES NATIONWIDE) **Tender Reference:** RFP-TELTECH-FWA-2026-001 **Project:** Nationwide 5G FWA Project (Band n50, 60 MHz TDD) **Operator:** Telecommunications Technology Solutions Incorporated (TelTech) **Author & Authority:** OLIVER TUNGOL *Instructions to Bidders: Complete all unit rates in USD (\$) and Philippine Pesos (PHP ₱ at baseline FX 55.00) using the official Excel BoQ file (06_RFP_Commercial_Price_Book_and_BoQ_Template.xlsx). All prices must be firm, fixed, and valid for 180 days. --- ###* **TABLE 1: LOT 1 — RAN HARDWARE & ACCESS EQUIPMENT (PER SITE BASIS) | Item Ref | Equipment Description | Technical Sizing / Model | Qty**

per Site | Unit Price (USD) [Bidder to Quote] | Total Price per Site (USD) | Remarks / Bidder Model |

RAN-HW-01	5G Baseband Unit (BBU)	19" subrack, dual -48V DC PSU, 4x 25G backhaul, 6x 25G eCPRI	1 set	[Bidder Quote]	[Formula = Qty * Rate]	Specify model & datasheet
RAN-HW-02	8T8R Remote Radio Unit (RRU)	Band n50 (1.5 GHz), 60 MHz IBW, 8x40W (320W total RF power)	3 sets	[Bidder Quote]	[Formula = Qty * Rate]	3 Sectors per site
RAN-HW-03	Macro High-Gain Antennas	8-port cross-pol, 17.5 dBi gain, AISG 2.0 internal eRET	3 sets	[Bidder Quote]	[Formula = Qty * Rate]	Or integrated AAU option
RAN-HW-04	RF Jumpers & Optical Patchcords	High-grade armored optical fiber, weatherproof boots, grounding kits	1 lot	[Bidder Quote]	[Formula = Qty * Rate]	All accessories included
RAN-HW-05	BBU Base Software & 5G Features	Band n50 carrier license, MU-MIMO, beamforming, sleep-mode energy saving	1 lot	[Bidder Quote]	[Formula = Qty * Rate]	Perpetual base software
SUBTOTAL	BTS Equipment Package per Site	(BBU x1, 8T8R RRU x3, Antennas x3, SW)	1 Site	—	[Sum of Items HW-01 to HW-05]	Total BTS Hardware
RAN-SVC-01	Site Survey, Engineering & TSSR	Tower audit, structural verification, drone LOS photo sweep, civil sketch	1 site	[Bidder Quote]	Field survey report delivery	

RAN-HW-01	5G Baseband Unit (BBU)	19" subrack, dual -48V DC PSU, 4x 25G backhaul, 6x 25G eCPRI	1 set	[Bidder Quote]	[Formula = Qty * Rate]	Specify model & datasheet	
RAN-SVC-02	Project Management & Mobilization	Dedicated rollout PMO, site acquisition coordination, milestone tracking	1 site	[Bidder Quote]	Dedicated PM team		
RAN-SVC-03	Installation & Commissioning (I&C)	Rigging, antenna mounting, cable routing, BBU wiring, initial power-up	1 site	[Bidder Quote]	Civil and telecom I&C		
RAN-SVC-04	Single Site Optimization (SSO/SSV)	Station testing, sector drive test, throughput call, ping, handover test	1 site	[Bidder Quote]	SSV acceptance sign-off		
RAN-SVC-05	Cluster Optimization Services	Cluster tuning, inter-site interference, tilt adjustments (per site share)	1 site	[Bidder Quote]	Cluster drive testing		
RAN-SVC-06	Central Warehousing & Logistics	Staging, asset tagging, kitting, indoor climate storage	1 site	[Bidder Quote]	Inventory management		
RAN-SVC-07	Delivery & Basic Hauling to Site	Regional hauling, rough-road 4x4 transport to rural barangay location	1 site	[Bidder Quote]	Complete site delivery		
SUBTOTAL	RAN Turnkey Services per Site	Full Turnkey Rollout & Commissioning Services	1 Site	[Sum of Items SVC-01 to SVC-07]	Total Turnkey Services	> Total Day-1 RAN Capex per Site (Equipment + Turnkey Services): [Sum of Hardware + Turnkey Services] > Phase 1 Program Total (1,000	Item

RAN-HW-01	5G Baseband Unit (BBU)	19" subrack, dual -48V DC PSU, 4x 25G backhaul, 6x 25G eCPRI	1 set	[Bidder Quote]	[Formula = Qty * Rate]	Specify model & datasheet	
						Sites): [Total per Site * 1,000] --- ### TABLE 3: LOT 2 — POWER SYSTEMS & PASSIVE INFRASTRUCTURE	
PWR-EQ-01	6.0 kW Modular DC Rectifier Plant	19" 3U subrack, 3 x 3.0 kW modules (N+1), SMPS controller	Co-loc / Greenfield	[Bidder Quote]	[Formula = USD * 55]		
PWR-EQ-02	48V / 150 Ah LiFePO4 Battery Bank	High-cycle Lithium Iron Phosphate with smart BMS (>= 4hrs backup)	Co-loc / Greenfield	[Bidder Quote]	[Formula = USD * 55]		
PWR-EQ-03	TowerCo DC Tap-Off & Metering Kit	DC tap enclosure, revenue Class 1.0 kWh meter, SPD Type 2	Leased TowerCo	[Bidder Quote]	[Formula = USD * 55]		
PWR-EQ-04	15 kVA Diesel Generator Set	Soundproof canopy, 500L–1000L tank, ATS, auto-start	Greenfield (Class T4)	[Bidder Quote]	[Formula = USD * 55]		
CIV-EQ-01	30m Monopole Tower & Foundation	Galvanized steel monopole, pile/pad foundation, compound fence	Greenfield (Class T4)	[Bidder Quote]	[Formula = USD * 55]		
CIV-EQ-02	Grounding Grid & Lightning System	Copper earth ring, Cadweld, ground rods, air terminal (<= 5\ Ohms)	Greenfield (Class T4)	[Bidder Quote]	[Formula = USD * 55]	--- ### TABLE 4: LOT 3 — 5G CORE NETWORK (5GCSA) & DATACENTER	Item
5GC-HW-01	Compute Server Nodes (2U)	Dual Xeon Gold / EPYC, 512GB RAM, NVMe	Hardware Purchase	[Bidder Quote]			

RAN-HW-01	5G Baseband Unit (BBU)	19" subrack, dual -48V DC PSU, 4x 25G backhaul, 6x 25G eCPRI	1 set	[Bidder Quote]	[Formula = Qty * Rate]	Specify model & datasheet
		(Dual Sites, N+1)				
5GC-HW-02	High-Throughput UPF Nodes	80 Gbps line-rate forwarding per node with DPDK (N+1 per DC)	Hardware Purchase	[Bidder Quote]		
5GC-HW-03	Datacenter Network Fabric	Spine/leaf switches (100GE / 25GE), perimeter firewalls, anti-DDoS	Hardware Purchase	[Bidder Quote]		
5GC-HW-04	Storage Array & Backup System	High-IOPS redundant enterprise NVMe storage arrays	Hardware Purchase	[Bidder Quote]		
5GC-SW-01	5GC Control Plane Software (1M)	AMF, SMF, PCF, UDM, UDR, AUSF, NRF, NSSF, CHF (1,000,000 Subs)	Perpetual License	[Bidder Quote]		
5GC-SW-02	Carrier-Grade NAT & DPI Software	CG-NAT deterministic port license + Layer-7 DPI bandwidth policing	Perpetual License	[Bidder Quote]		
5GC-SW-03	CaaS / Kubernetes Orchestration	Enterprise Kubernetes platform & container management	Perpetual / Annual	[Bidder Quote]		
5GC-SVC-01	Integration, IOT with RAN & UAT	End-to-end design, commissioning, RAN interoperability testing	Professional Services	[Bidder Quote]		

| **SUBTOTAL | 5G Core Year 1 Turnkey Package** | *Dual Datacenter Active-Active Setup (1M Subs)* | **Turnkey Initial**
| **[Sum of Items 5GC-01 to SVC-01]** | *Incremental Capacity Expansion License (Tiers above 1,000,000*
Subscribers):

- *Subscribers 1,000,001 to 2,000,000*: \ [Bidder to Quote]\$ per registered subscriber.
- *Subscribers 2,000,001 to 3,000,000*: \ [Bidder to Quote]\$ per registered subscriber.
- *Annual Software Maintenance & Support (AMC)*: [Bidder to Quote] % per annum. --- ### TABLE 5: LOT 4 —
DIGITAL OSS/BSS & FIELD FORCE AUTOMATION | Item Ref | OSS/BSS Functional Module | Detailed Functional
Scope | License Type | Total Year 1 Price (USD) [Bidder Quote] |

BSS-01	Converged CRM & Product Catalog	Omnichannel customer onboarding, product catalog (4 FWA tariffs)	Perpetual / Enterprise	[Bidder Quote]
BSS-02	Real-Time Balance & Rating Engine	CHF adapter, real-time prepaid quota metering, zero-balance redirect	Perpetual License	[Bidder Quote]
BSS-03	Payment Channel Integration	Native API integrations: GCash, Maya, 7-Eleven CLiQQ, Bayad Center	Integration Module	[Bidder Quote]
BSS-04	Billing & Invoicing Engine	Postpaid billing engine for SME and enterprise anchor tenants	Perpetual License	[Bidder Quote]
OSS-01	Field Force Management (FMS)	Algorithmic automated dispatch handling 3,500 daily work orders	Enterprise License	[Bidder Quote]
OSS-02	Mobile Technician Android App	Barcode scan, GPS dispatch, live RF RSRP/SINR validation tool	Unlimited Users	[Bidder Quote]
OSS-03	CPE Warehouse Logistics & Reverse WMS	Serialized asset lifecycle tracking, repossession & refurbishment module	Enterprise License	[Bidder Quote]
OSS-04	Network Performance & Fault OSS	Multi-vendor RAN/Core EMS integration, alarm	Enterprise License	[Bidder Quote]

BSS-01	Converged CRM & Product Catalog	Omnichannel customer onboarding, product catalog (4 FWA tariffs) correlation, RCA	Perpetual / Enterprise	[Bidder Quote]	--- ### TABLE 6: LOT 6 — MANAGED SERVICES & 24/7 NETWORK OPERATIONS (ANNUAL OPEX)		Item Ref	Managed Operations Domain
OSS-05	Regulatory BI & NTC Dashboard	Automated QoS speed compliance reporting to NTC	Included	[Bidder Quote]				
OSS-SVC	Systems Integration & UAT Launch	End-to-end configuration, test scripts, launch support	Professional Services	[Bidder Quote]				
SUBTOTAL	OSS/BSS Turnkey Year 1 Package	Full Stack Suite Dimensioned for 1M Subscribers	Turnkey Initial	[Sum of Items BSS-01 to OSS-SVC]				

MS-OP-01	24/7 Tier-2/Tier-3 Central NOC	Active surveillance, incident triage, vendor escalation	Per Site / Month	[Bidder Quote]	[PHP / 55]	[Monthly USD * 12]
MS-OP-02	Field First Line Maintenance (FLM)	24/7 nationwide emergency restoration, MTTR <= 2-4hrs	Per Site / Month	[Bidder Quote]	[PHP / 55]	[Monthly USD * 12]
MS-OP-03	Scheduled Preventive Maintenance	Quarterly site inspection, power, battery, grounding	Per Site / Month	[Bidder Quote]	[PHP / 55]	[Monthly USD * 12]
MS-OP-04	Spare Parts Logistics & Warehousing	Regional buffer stock, NBD replacement delivery	Per Site / Month	[Bidder Quote]	[PHP / 55]	[Monthly USD * 12]

BSS-01		Converged CRM & Product Catalog	Omnichannel customer onboarding, product catalog (4 FWA tariffs)	Perpetual / Enterprise	[Bidder Quote]		
MS-OP-05	Continuous Network RF Optimization	Routine KPI monitoring, neighbor tuning, interference mgmt	Per Site / Month	[Bidder Quote]	[PHP / 55]	[Monthly USD * 12]	
TOTAL	Active Network O&M Fee	All inclusive active telecom managed operations	Per Site / Month	[Sum of OP-01 to OP-05]	[Sum of USD]	[Annual Total USD]	--- ### TABLE LOT 7 - CUSTC PREMI: EQUIPI (CPE)
CPE-01	Indoor 5G High-Gain FWA Router	Band n50 native, 8-10 dBi internal antenna, Wi-Fi 6 AX1800, 2.5GE port, TR-069	350,000 units	[Bidder Quote]	[Formula = Qty * Rate]		
CPE-02	Outdoor 5G High-Gain FWA Unit	Band n50, IP67 enclosure, 12 dBi directional antenna, PoE injector, 2.5GE LAN	15,000 units	[Bidder Quote]	[Formula = Qty * Rate]		
CPE-SVC	Staging, Pre-Config & SIM-Locking	APN, firmware customization, PLMN locking, customer packaging	350,000 units	[Bidder Quote]	[Formula = Qty * Rate]	--- ### TABLE 8: MANDATORY VENDOR FINANCING FACILITY (FORM D) The Bidder must submit a formal, legally binding financing term sheet with the following parameters:	Financ Lever

BSS-01 Converged CRM & Product Catalog Omnichannel customer onboarding, product catalog (4 FWA tariffs) Perpetual / Enterprise [Bidder Quote]		
Financing Structure	OEM-backed Buyer's Credit or Export Credit Agency (ECA) facility	[Bidder to Specify Facility Structure]
Financing Quantum	Minimum 80% of Eligible Capital Expenditure (Lots 1, 3, Turnkey Services)	[Bidder to State % Financed]
Equity Downpayment	Maximum 20% Downpayment upon contract signing	[Bidder to State Downpayment %]
Facility Tenor	4 to 5 Years total facility duration	[Bidder to State Tenor in Months]
Principal Grace Period	12 to 18 Months grace period from first disbursement	[Bidder to State Grace Period]
Indicative Interest Rate	Fixed commercial rate (or SOFR + competitive margin)	[Bidder to State Fixed Rate or Spread]
Security / Collateral	Pledge on network equipment assets without restrictive operational covenants	[Bidder to Specify Security Terms]

| **Repayment Frequency** | Semi-annual or quarterly principal and interest amortizations | [Bidder to Specify Payment Cadence] | ---

(End of Appendix B — Commercial Proposal & Pricing Templates)